Reg. No. | | | | | | | | | | | [12ME74]

SEVENTH SEMESTER B.E EXAMINATION, DECEMBER-2015

MECHATRONICS

Time: 3 Hours **Maximum Marks:70**

Instructions: Answer any **FIVE** full Questions.

1. a. State the objectives of mechatronics. Bring out the advantages and disadvantages of a mechatronic system. **06**

b. Explain the mechatronics based engine management system with a block diagram. **08**

2. a. State the principle of operation of transducers and highlight their differences with sensors. **06**

b. Explain how level of a liquid in a tank is measured with the help of a potentiometer. **08**

3. a. What are switching devices, relay, solenoids? How many types are there in each. **06**

b. What is a valve and how are they classified? Sketch and briefly explain the working of a rotary four way D.C valve used is hydraulic system. **08**

4. a. Elucidate the terms **06**

i. Mechanism

ii. Kinetic chain with an example

b. Differentiate between **08**

i. FET and MOSFET

ii. Stepper motor and servomotor

5. a. Write a note on **06**

i. Encoders

ii. Timers

iii. Counters

b. With an example distinguish between open loop and closed loop systems and list their advantages and disadvantages. **08**

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6. a. Describe the following process control systems **06**

i. PLC **ii.** SCADA **iii.** HMI

b. Explain the structure of a microprocessor and draw the architecture of 8085 microprocessor. **08**

7. a. Specify different techniques that can be used to detect faults. Explain any two of them in brief. **06**

b. Enumerate fault finding techniques that are used with microprocessor based system and systematic fault location method. **08**

8. a. Explicate how the mechatronics designs are different from Traditional designs. **06**

b. Describe any two applications of mechatronics in **08**

i. Manufacturing

ii. Bionics

iii. Avionics